

Итерация 3

Проект 20 - ML- классификатор водоёмов на основе спутниковых СНИМКОВ И ОТКРЫТЫХ данных

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План на текущую итерацию

- Добавить тестирование
- Улучшение работы модели распознавание болот

Результаты текущей итерации

- Тесты
 - water-detection
 - get_satellite_image
 - water-info
 - cv_integrated_water_classifier
 - get_eutrophication_stats
- Реализован автоматический запуск тестов в github actions
- Selenium тест

Пример работы github actions

All workflows					Filter workflow runs
Showing runs from all workflows					
14 workflow runs					
		Workflow	Event	Status	Branch
14 workflow runs					Actor
✓	Merge pull request #77 from moevm/issue-54	main		10 minutes ago	...
CI #14: Commit 09a65c0 pushed by Margo63				2m 20s	
✓	add workflow + fix requirements + speed build	issue-54		Apr 25 at 3:04 PM GMT+3	...
CI #13: Pull request #77 synchronize by nikitakozmin				1m 44s	
✓	add workflow + fix requirements + speed build	issue-54		Apr 25 at 3:00 PM GMT+3	...
CI #12: Pull request #77 synchronize by nikitakozmin				43s	

Пример запуска одного теста

```
(base) margarita@laptop mse1h2026-water % pytest tests/test_selenium.py
===== test session starts =====
platform darwin -- Python 3.11.7, pytest-9.0.2, pluggy-1.6.0 -- /usr/local/anaconda3/bin/python3
rootdir: /Users/margarita/UPRP0/project/mse1h2026-water
configfile: pytest.ini
plugins: mock-3.15.1, repeat-0.9.4, colcon-core-0.20.1, anyio-4.2.0, rerunfailures-16.1, cov-7.0.0
collected 1 item

tests/test_selenium.py::test_streamlit_coordinates PASSED [100%]/
usr/local/anaconda3/lib/python3.11/site-packages/coverage/control.py:958: CoverageWarning: No data was collected. (no-data-collected); see https://coverage.readthedocs.io/en/7.13.5/messages.html#warning-no-data-collected
  self._warn("No data was collected.", slug="no-data-collected")

===== tests coverage =====
TOTAL 0%
===== 1 passed in 103.12s (0:01:43) =====
```

Пример запуска всех тестов

```
(base) margarita@laptop mse1h2026-water % pytest
===== test session starts =====
platform darwin -- Python 3.11.7, pytest-9.0.2, pluggy-1.6.0 -- /usr/local/anaconda3/bin/python3
rootdir: /Users/margarita/UPRPO/project/mse1h2026-water
configfile: pytest.ini
testpaths: tests
plugins: mock-3.15.1, repeat-0.9.4, colcon-core-0.20.1, anyio-4.2.0, rerunfailures-16.1, cov-7.0.0
collected 24 items

tests/test_cv_integrated_water_classifier.py::test_cv_classifier_success PASSED [ 4%]
tests/test_cv_integrated_water_classifier.py::test_cv_classifier_calls_download_correct PASSED [ 8%]
tests/test_cv_integrated_water_classifier.py::test_cv_classifier_invalid_payload PASSED [ 12%]
tests/test_cv_integrated_water_classifier.py::test_cv_classifier_passes_download_result_to_classifier PASSED [ 16%]
tests/test_cv_integrated_water_classifier.py::test_cv_classifier_does_not_call_classifier_when_download_fails PASSED [ 20%]
tests/test_cv_integrated_water_classifier.py::test_cv_classifier_download_called_before_classifier PASSED [ 25%]
tests/test_cv_integrated_water_classifier.py::test_cv_classifier_calls_initialize_ee PASSED [ 29%]
tests/test_get_eutrophication_stats.py::test_get_eutrophication_stats_returns_payload PASSED [ 33%]
tests/test_get_eutrophication_stats.py::test_get_eutrophication_stats_default_params PASSED [ 37%]
tests/test_get_eutrophication_stats.py::test_get_eutrophication_stats_invalid_latitude PASSED [ 41%]
tests/test_get_eutrophication_stats.py::test_get_eutrophication_stats_invalid_longitude PASSED [ 45%]
tests/test_get_eutrophication_stats.py::test_get_eutrophication_stats_calls_initialize_ee_once PASSED [ 50%]
tests/test_get_satellite_image.py::test_get_satellite_image_returns_metadata PASSED [ 54%]
tests/test_get_satellite_image.py::test_get_satellite_image_defaults PASSED [ 58%]
tests/test_get_satellite_image.py::test_get_satellite_image_invalid_lat PASSED [ 62%]
tests/test_get_satellite_image.py::test_get_satellite_image_invalid_lon PASSED [ 66%]
tests/test_get_satellite_image.py::test_get_satellite_image_calls_initialize_ee PASSED [ 70%]
tests/test_selenium.py::test_streamlit_coordinates PASSED [ 75%]
tests/test_water_detector.py::test_water_detector_payload_from_service PASSED [ 79%]
tests/test_water_detector.py::test_water_detector_default_params PASSED [ 83%]
tests/test_water_detector.py::test_water_detector_invalid_latitude PASSED [ 87%]
tests/test_water_detector.py::test_water_detector_non_integer_radius PASSED [ 91%]
tests/test_water_info.py::test_get_water_info_success PASSED [ 95%]
tests/test_water_info.py::test_get_water_info_default_params PASSED [100%]

===== tests coverage =====
TOTAL 98%
===== 24 passed in 118.78s (0:01:58) =====
```

План на следующую итерацию

- Улучшение работы модели распознавание болот
- Добавить ООП
- Интерпритация риска
- Расширение selenium тестов (не правильный ввод координат, загрузка бэка, ошибка подгрузки изображения)